

What the Research Says About Your Metal Risk

A plain-language briefing for AMC Company

This briefing explains what a multi-year study of precious-metals price behaviour found, what it failed to find, and what both mean for how AMC buys and carries metal. It is written to be read start to finish in about fifteen minutes, with no statistical background assumed.

Prepared for: The owner, AMC Company

Scope: Gold, silver, platinum, palladium — scrap intake and coin trading

Price data behind it: 56,256 daily records, 01 Jan 2007 to 22 May 2026

News data behind it: 139,904,912 articles

Model runs logged: 55

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If you read nothing else

- **One reliable warning signal exists.** When the Fed surprises the market by sounding tougher on rates, metal prices fall over the next week or so — silver about twice as hard as gold. This is the single result that survived every test we threw at it.
- **Most popular market wisdom did not survive.** “Weak dollar lifts metals” and “geopolitical tension lifts gold” both failed. Knowing which rules of thumb are false is worth as much as the one that is true, because it stops you acting on them.
- **News and sentiment tracking does not predict prices.** We tested this properly, with the pass mark set in advance, on 139.9 million articles. It failed, and on fresh data it actively made forecasts worse. We recommend not paying for a sentiment feed.
- **Simple beats sophisticated.** On data the models had never seen, decades-old statistical methods forecast better than the machine learning. We are recommending the simpler tool.
- **The biggest gap is your own books.** Everything above describes the market. Converting it into dollars of risk on YOUR inventory requires your transaction ledger, which we do not yet have. That is the single highest-value thing you can provide.

THE HONEST HEADLINE

This programme produced one durable trading-relevant fact, several useful refutations, and a clear-eyed account of what it could not establish. If a research report ever tells you everything worked, be suspicious of it.

What holds up

Each finding below survived multiple independent methods. The caveat printed with each one is not boilerplate — it is the specific condition under which the finding stops being true.

A hawkish Fed surprise pushes metal prices down for about a week

When the Federal Reserve surprises the market by sounding tougher on interest rates than expected, gold, silver and platinum reliably fall over the following week. This is the one result we would put real weight behind.

The numbers. Five trading days after a hawkish surprise: gold -1.5% , silver -3.0% , platinum -1.7% , palladium -1.6% (palladium not statistically reliable). Over twenty days the moves roughly double: gold -1.8% , silver -3.7% , platinum -3.0% .

How confident. Highest. Three independent methods agree, including one that never looks at Fed meeting dates at all. Holds in all three time periods we split the data into. Shuffling the event dates at random destroys the effect, as it should.

CAVEAT THAT TRAVELS WITH THIS

The effect has weakened by roughly $2.3\times$ since the money-printing era — a surprise today moves prices less than the same surprise in 2010. Based on 35 events. And critically: we could NOT re-test this on recent data, because the dataset measuring Fed surprises stops in December 2023. The entire 2024–2026 rate-cutting cycle is untested.

Metals punish bad Fed news far harder than they reward good Fed news

A tough Fed surprise hurts. A friendly Fed surprise barely helps. Metal behaves like insurance: it reacts sharply when the reason to hold it is threatened, and only mildly when that reason is reaffirmed.

The numbers. Hawkish surprise: gold -1.4% over five days. Dovish (friendly) surprise: gold $+0.6\%$, and not statistically distinguishable from zero. No dovish result for any metal reached significance.

How confident. High. Both main methods agree on the asymmetry.

CAVEAT THAT TRAVELS WITH THIS

Our sample period (2010–2023) was tilted toward tightening cycles, so some of the muted upside may reflect that easy money was already priced in rather than a permanent behavioural rule.

Silver is your most volatile exposure; gold is your steadiest

When a Fed shock hits, silver swings hardest, platinum next, gold and palladium least. This ranking held up across every method and every time period we tested — it is one of the most stable patterns in the work.

The numbers. Silver moves roughly twice as far as gold on the same shock (−3.0% vs −1.5% at five days).

How confident. High. Ordering preserved across both estimators and all three eras.

CAVEAT THAT TRAVELS WITH THIS

Palladium is the unreliable member of the group: its response to Fed news collapses to near zero after 2015, almost certainly because the 2018–2022 industrial supply squeeze overwhelmed anything the Fed was doing.

What failed, and why that is worth money to you

A study that only reports what worked is selling you something. These are the ideas we tested and rejected. Each one represents money or attention you no longer need to spend.

News and sentiment data does not help predict prices — it hurts

We collected 139.9 million news articles and built market-mood categories from them. They did not improve volatility forecasts. On fresh data they made forecasts significantly WORSE. We set the pass mark before running the test, and the test failed it.

The numbers. The pre-set bar was a 1.0% accuracy improvement and wins in 60% of test periods. Actual: 0.37% WORSE, winning 4 of 11. On the untouched hold-out data, news-regime and sentiment features both degraded accuracy by a statistically significant margin.

How confident. High, and unusually clean: the standard was fixed in advance, then independently reconfirmed on data never used in development.

IMPORTANT NUANCE

This means news has no value for FORECASTING. It does not mean news is worthless — the same categories are useful for explaining why a past move was large. Prediction and explanation are different jobs.

The machine-learning model did not beat old-fashioned statistics

On data the models had never seen, two decades-old statistical methods produced lower forecast errors than the machine-learning model. We are reporting the simpler tools as the better ones.

The numbers. Forecast error on the hold-out: classical methods 0.127 and 0.131, machine learning 0.140. A naive guess was worst at 0.186 — so the modelling is doing something, just not better than the simple version.

How confident. Moderate. The test window contains only about two or three genuinely independent observations, which is too few to settle the question.

IMPORTANT NUANCE

No model was tuned, by design, to keep the comparison fair. A tuned machine-learning model might do better — we have not tested that.

“A weak dollar is good for metals” did not survive testing

This common rule of thumb came out backwards in our data, and then behaved even more strangely on fresh data. We recommend not trading on it in this simple form.

The numbers. Sharp dollar declines were followed by metal prices FALLING, not rising (platinum -1.6%, palladium -2.2%). On fresh data the direction repeated but the size was 5 to 90 times too large — a warning sign, not a confirmation.

How confident. We are confident the simple version is broken, and we understand why: big dollar moves cluster inside panics, so the measure picks up the panic rather than the currency.

IMPORTANT NUANCE

A properly isolated dollar effect DOES exist and points the textbook way. The problem is the crude trigger, not the underlying economics.

Geopolitical risk headlines are not a usable gold signal

The belief that spiking geopolitical tension reliably lifts gold did not hold. The index we used cannot tell a genuine crisis from a merely noisy news week.

The numbers. Across 207 spike events, gold's five-day move was +0.05% — indistinguishable from nothing. On fresh data the signal was a coin flip: two of four metals moved the predicted way.

How confident. High that the signal as constructed is unusable.

IMPORTANT NUANCE

Flight-to-safety is real — we can see it when we identify crises by market behaviour instead of headline counts. The measuring instrument was at fault.

The buy-spread tool, and why you cannot use it yet

The practical product of this work is a calculator that answers your daily question: *how far below spot must I buy so that a bad week does not turn this lot into a loss?* It works from the exit price you can realise, minus a cushion for how far the metal could fall while you hold it, minus your cost of carrying it.

DO NOT QUOTE THESE NUMBERS TO A CUSTOMER

The figures below are placeholders, not prices. Every one of them rests on an assumed two-week holding period and a placeholder exit price, because your ledger and refiner settlements are not yet loaded. They are shown so you can see the machinery working and judge whether the shape is right — they are deliberately too conservative, and using them as quotes would cost you deals.

Metal	Spot price	Maximum buy	% of spot	Discount
Gold	\$4,523.20	\$4,072.69	90.0%	10.0%
Palladium	\$1,360.30	\$1,087.00	79.9%	20.1%
Platinum	\$1,939.70	\$1,590.50	82.0%	18.0%
Silver	\$76.20	\$61.46	80.7%	19.3%

Computed 22 May 2026, the last date with complete price data. Placeholder values — see the caution above.

What each number currently rests on

- **Gold** uses a forward-looking price-swing estimate taken from the gold options market.
- **Silver, platinum and palladium** use a backward-looking estimate taken from each metal's own recent declines — no equivalent options market exists for them.
- Bad-case size uses a textbook bell-curve rule, which understates genuine panics.
- Holding period is ASSUMED at 10 trading days. Not measured from your books, because we do not have them yet.
- Financing cost counts the Treasury rate only — not your insurance, labour, or shop overhead.
- Exit price is a placeholder haircut, not your actual refiner payable or dealer bid.

Read that list as a to-do rather than a disclaimer. Every item on it tightens once real data replaces the placeholder — and a tighter floor means you can bid more competitively without taking on more risk.

What would make these numbers real

What we need	What it fixes	Where it comes from
Your scrap ledger — purchase and disposal dates, weights, fineness, price paid, proceeds	Replaces the assumed two-week holding period with your actual one. This is the single largest source of excess conservatism.	Your bookkeeping system. Template and importer are already built and tested.
Refiner settlement statements	Replaces the placeholder exit price with your true payable.	Your refiner invoices.
Your coin buy and sell records	Measures the premium you actually realise, rather than a published benchmark that may not reflect your market.	Your bookkeeping system.

What we need from you

NOTHING RECEIVED YET — THIS IS THE BOTTLENECK

We hold zero rows of your transaction data. The import tool, the file format, and the validation checks are all built and tested; they are waiting on a first export from your bookkeeping. Until then every dollar-denominated figure in this briefing rests on an assumption rather than on your business.

Three practical points about that data. **First, it never leaves the machine.** Your books are held on one local computer, are never uploaded to any cloud service, and are excluded from any copy that goes off-site. **Second, it does not need to be clean.** The importer checks every row and reports every problem at once, and corrected files can simply be re-imported. **Third, a partial history is still useful** — even a few hundred closed lots would replace the assumed holding period with a measured one.

What you would get back, quickly

- How long each metal actually sits in your inventory, which is the number that sets how wide your buying spread must be.
- Your realised margin per lot, by metal and by lot size.
- The premium you genuinely capture on coins, by product — measured, not benchmarked.
- A dollar figure for how much a bad week costs your current book, which is the number that tells you whether hedging is worth its cost.

What this work cannot tell you

- **It does not predict prices.** No part of this study forecasts the direction of gold. That is not a shortcoming of the effort; it is what the evidence supports.
- **The main finding is untested on the last two years.** The dataset measuring Fed surprises ends in December 2023, so the whole 2024–2026 rate-cutting cycle is unexamined. This is the most valuable gap to close.
- **Palladium is the weakest case throughout.** Its industrial supply story overwhelms the financial patterns that fit the other three metals. Treat palladium conclusions with extra caution.
- **The hold-out test was small.** About two or three genuinely independent observations. It was enough to rule things out, not enough to confirm fine distinctions.
- **Some data collection is paused on legal grounds.** Several automated price and premium feeds were found to prohibit commercial use, so they were stopped and 171,245 already-collected rows were quarantined out of all analysis pending a licence. Nothing in this briefing relies on them.

Terms used in this briefing

Term	What it means
Hawkish surprise	The Fed signals tighter money — higher rates, or rates staying high longer — than the market expected that morning.
Dovish surprise	The opposite: easier money than expected.
Float	How long you own metal between buying it and disposing of it. Risk grows with the square root of this period, so doubling it raises required cushion by about 41%, not 100%.
Hold-out test	Data locked away and never looked at while building the models, then used once at the end as a closed-book exam.
Pre-registered test	Writing down the pass mark before running the test, so the result cannot be rationalised afterwards.
Spot	The live market price per troy ounce of pure metal.
Payable	The share of metal value your refiner actually pays you.

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